

CHAPTER 10

Amines

Preparation of Amines

- Reaction of propanamide with ethanolic sodium hydroxide and bromine will give (2020-Covid)
 - Methylamine
 - Propylamine
 - Aniline
 - Ethylamine
- Method by which Aniline cannot be prepared is: (2015 Re)
 - Potassium salt of phthalimide treated with chlorobenzene followed by hydrolysis with aqueous NaOH solution
 - Hydrolysis of phenylisocyanide with acidic solution
 - Degradation of benzamide with bromine in alkaline solution
 - Reduction of nitrobenzene with H_2/Pd in ethanol

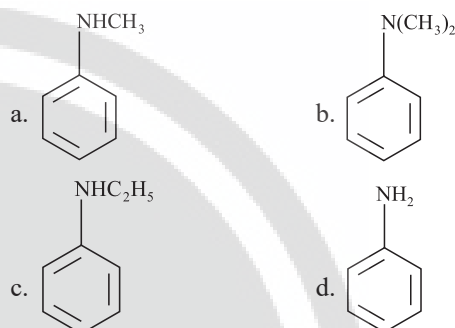
Physical & Chemical Properties of Amines

- Given below are two statements: (2022)
 Statements I : Primary aliphatic amines react with HNO_2 to give unstable diazonium salts.
 Statements II : Primary aromatic amines react with HNO_2 to form diazonium salts which are stable even above 300 K.
 In the light of the above Statements, choose the most appropriate answer from the options given below:
 - Statements I is incorrect but Statements II is correct.
 - Both Statements I and Statements II are correct
 - Both Statements I and Statements II are incorrect.
 - Statements I is correct but Statement II is incorrect.

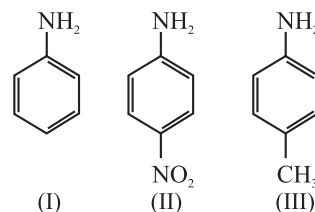
- Identify the compound that will react with Hinsberg's reagent to give a solid which dissolves in alkali. (2021)

- $CH_3-CH_2-NH-CH_3$
- $CH_3-CH_2-NH_2$
- $CH_3-CH_2-N(CH_3)-CH_2-CH_3$
- $CH_3-CH_2-NO_2$

- Which of the following amine will give the carbylamine test? (2020)



- The correct order of the basic strength of methyl substituted amines in aqueous solution is: (2019)
 - $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N$
 - $(CH_3)_3N > CH_3NH_2 > (CH_3)_2NH$
 - $(CH_3)_3N > (CH_3)_2NH > CH_3NH_2$
 - $CH_3NH_2 > (CH_3)_2NH > (CH_3)_3N$
- Nitration of aniline in strong acidic medium also gives m-nitroaniline because: (2018)
 - In spite of substituents nitro group always goes to only m-position.
 - In electrophilic substitution reactions, amino group is meta directive.
 - In acidic (strong) medium, aniline is present as anilinium ion.
 - In absence of substituents nitro group always goes to m-position.
- The correct increasing order of basic strength for the following compounds is: (2017-Delhi)

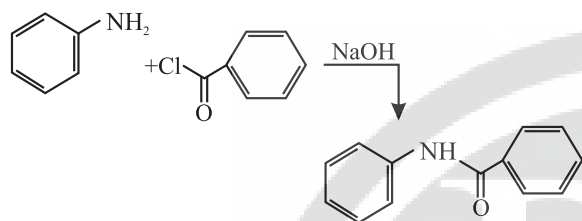


- $II < I < III$
- $II < III < I$
- $III < I < II$
- $III < II < I$

9. The correct statement regarding the basicity of arylamines is: (2016 - I)

- Arylamines are generally more basic than alkylamines, because the nitrogen atom in arylamines is sp -hybridised.
- Arylamines are generally less basic than alkylamines because the nitrogen lone pair electrons are delocalized by interaction with the aromatic ring π electrons system.
- Arylamines are generally more basic than alkylamines because the nitrogen lone pair electrons are not delocalised by interaction with the aromatic ring π electron system.
- Arylamines are generally more basic than alkylamines because of aryl group

10. The following reaction



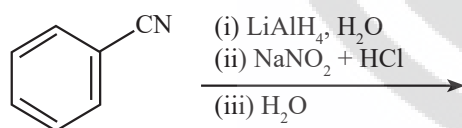
is known by the name:

[OS] (2015 Re)

- Schotten-Baumen reaction
- Friedel-Craft's reaction
- Perkin's reaction
- Acetylation reaction

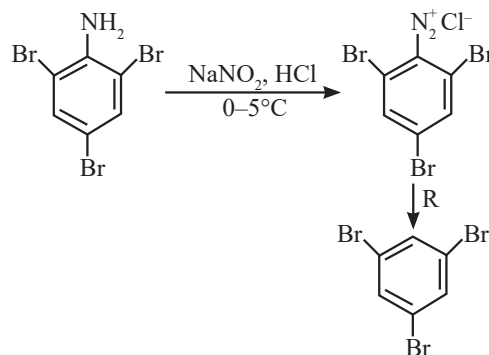
Diazonium Salt (Preparation, Physical & Chemical Properties)

11. The product formed from the following reaction sequence is (2022)



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12. The reagent 'R' in the given sequence of chemical reaction is: (2021)

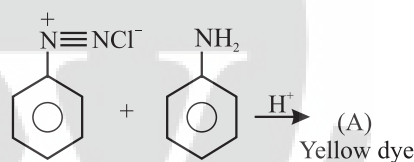


- $\text{CH}_3\text{CH}_2\text{OH}$
- HI
- CuCN/KCN
- H_2O

13. A given nitrogen-containing aromatic compound A reacts with Sn/HCl , followed by HNO_2 to give an unstable compound B. B, on treatment with phenol, forms a beautiful coloured compound C with the molecular formula $\text{C}_{12}\text{H}_{10}\text{N}_2\text{O}$. The structure of compound A is: (2016 - II)

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14. In the following reaction, the product (A) is: (2014)

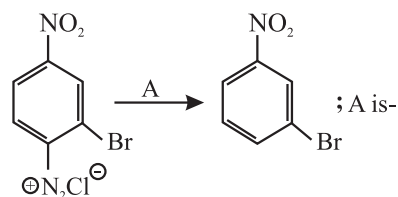


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15. Which of the following will be most stable diazonium salt RN_2^+X^- ? (2014)

- $\text{C}_6\text{H}_5\text{N}_2^+\text{X}^-$
- $\text{CH}_3\text{CH}_2\text{N}_2^+\text{X}^-$
- $\text{C}_6\text{H}_5\text{CH}_2\text{N}_2^+\text{H}^-$
- $\text{CH}_3\text{N}_2^+\text{X}^-$

16. In the reaction:



- $\text{HgSO}_4/\text{H}_2\text{SO}_4$
- Cu_2Cl_2
- H_3PO_2 and H_2O
- $\text{H}^+/\text{H}_2\text{O}$

[illegible]